



Hardware Architecture

Optimized for ATmega328P Zero PCB Assembly

!IMPORTANT

Status Overview:

- INPUT_PULLUP: Active
- Logic Level: **3.3V** Interface
- Board Target: Arduino Nano / ATmega328P



Core Design Concepts

1. Internal Pull-Up Resistors

Benefit: Saves $4 \times 10\text{k}\Omega$ External Resistors

Instead of wiring external pull-down resistors to ground, the firmware activates the ATmega328P's internal pull-up resistors.

- **Idle State:** Pin is held at HIGH ($\approx 5\text{V}$ internally).
- **Pressed State:** Button connects pin directly to GND, reading LOW.
- **Implementation:** `pinMode(BTN_PIN, INPUT_PULLUP);`
- **Impact:** Reduces perfboard clutter and improves noise immunity.

2. Display Logic Level Shifting

Benefit: Prevents PCD8544 Controller Damage

The Nokia 5110 operates strictly at **3.3V**.

- **Signal Protection:** Series resistors or voltage dividers are used on SCLK, DIN, DC, and CE lines to drop 5V Arduino signals to **3.3V** logic thresholds.
- **Contrast (V0):** Requires precise $\approx 3\text{V}$. A $10\text{k}\Omega$ potentiometer is recommended to prevent a "white screen" effect.

3. Buzzer Current Limiting

Benefit: $1\text{k}\Omega$ GPIO Protection

The passive buzzer is driven by PWM pin D6.

- **Function:** Prevents GPIO overcurrent (limit $\approx 40\text{mA}$ per pin).
- **Acoustics:** Dampens coil resonance spikes while maintaining pressure for frequencies between 200Hz and 2000Hz .



Updated Wiring Table

Based on Firmware v3_full

Module	Terminal	Nano Pin	Logic Type
Nokia 5110	RST	D7	3.3V (Shifted)
	CE	D8	3.3V (Shifted)
	DC	D9	3.3V (Shifted)
	DIN	D10	3.3V (Shifted)
	CLK	D11	3.3V (Shifted)
	VCC	3.3V	Direct 3.3V Rail
D-Pad	UP	D2	INPUT_PULLUP
	DOWN	D3	INPUT_PULLUP
	RIGHT	D4	INPUT_PULLUP
	LEFT	D5	INPUT_PULLUP
Audio	Passive (+)	D6	Series 1kΩ

⚡ Power & Ground Strategy

CRITICAL SAFETY CHECK

Never power the Nokia 5110 from the Nano's 5V pin. The PCD8544 controller maximum voltage is 3.6V. Connecting to 5V will permanently destroy the LCD.

Assembly Tips

- Common GND Rail:** Solder a thick bus wire along one edge for all GND connections.
- Decoupling:** Add a 100nF ceramic capacitor across 5V and GND near the Nano to filter buzzer noise.
- Debounce:** Mechanical bounce is handled via firmware (180ms delta); hardware RC filters are not required.